

Coin exchange program practical

```
#include <iostream>
#include <vector>
#include <climits>
using namespace std;

int main() {
    int n, amount;

    cout << "Enter number of coins: ";
    cin >> n;

    vector<int> coins(n);

    cout << "Enter coin denominations: ";
    for (int i = 0; i < n; i++) {
        cin >> coins[i];
    }

    cout << "Enter amount: ";
    cin >> amount;

    // DP array
    vector<int> dp(amount + 1, INT_MAX);

    // Base case
    dp[0] = 0;

    // Dynamic Programming
```

```

for (int i = 1; i <= amount; i++) {
    for (int j = 0; j < n; j++) {

        if (coins[j] <= i && dp[i - coins[j]] != INT_MAX) {

            int newCoins = dp[i - coins[j]] + 1;

            if (newCoins < dp[i]) {
                dp[i] = newCoins;
            }
        }
    }
}

// Check result
if (dp[amount] == INT_MAX) {
    cout << "Amount cannot be formed." << endl;
}
else {
    cout << "Minimum number of coins = " << dp[amount] << endl;
}

return 0;
}

```

Output:

Enter number of coins: 4

Enter coin denominations: 1 5 6 9

Enter amount: 11

Minimum number of coins = 2

===== Code Execution Sucessful =====